

Estimating Active Dimension of Training Dynamics from One Scalar Log

with an Application to Grokking

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One log can reveal low-dimensional recurrent dynamics.

On the grokking runs studied here, it gives a transition signal; stored trajectories reveal a reversible change in covariance effective rank.

01 Define the quantity

Can a scalar training log reveal how many independent components a recurrent parameter trajectory explores?

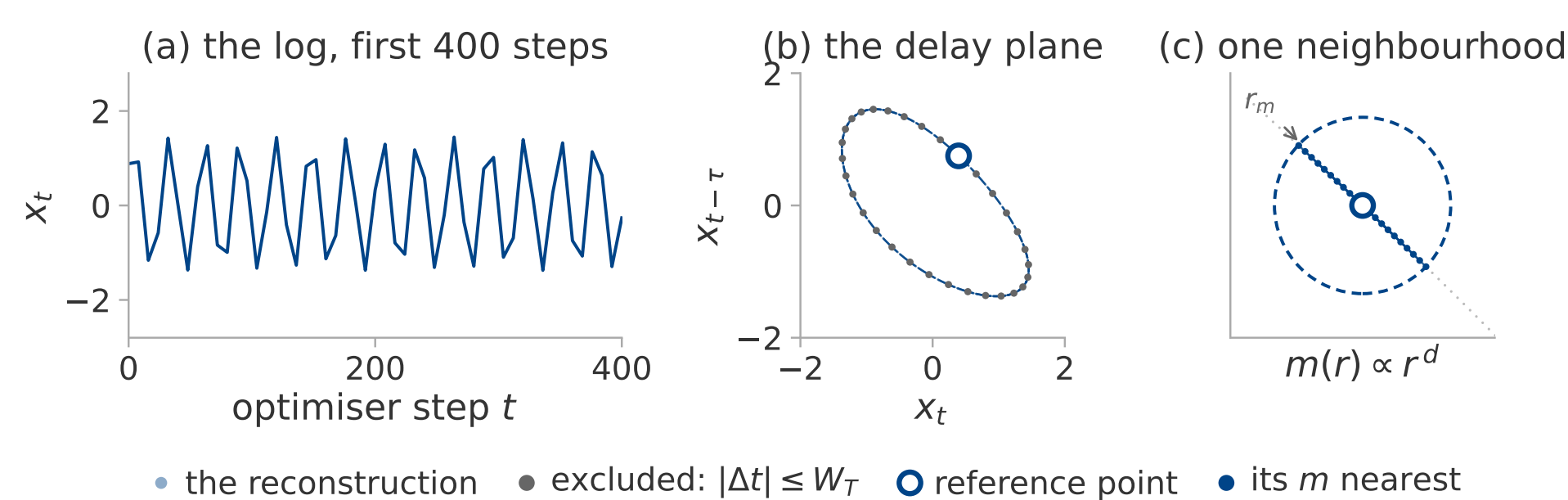
Active dimension is the local dimension of the occupation measure μ_R of a regime R :

$$d_{\text{act}}(R) = \lim_{\varepsilon \downarrow 0} \frac{\log \mu_R(B(c, \varepsilon))}{\log \varepsilon} \quad \text{for } \mu_R\text{-almost every } c.$$

Here c denotes parameter coordinates and $B(c, \varepsilon)$ a ball of radius ε . We assume the measure has the same local dimension almost everywhere. On a smooth occupied manifold with positive density, this counts local coordinates. Finite data resolve only finite scales.

Available	Directions allowed by the parameter subspace.
Functional	Rank of the output Jacobian on a fixed probe set.
Active	Dimension of the recurrent occupation measure.
Effective rank	Spread of trajectory variance across directions.

02 Reconstruct, then count neighbours



One scalar record, its delay reconstruction, and a neighbourhood. The loop is visited repeatedly.

$$y_t = (x_t, x_{t-\tau}, \dots, x_{t-(E-1)\tau}) \in \mathbb{R}^E,$$

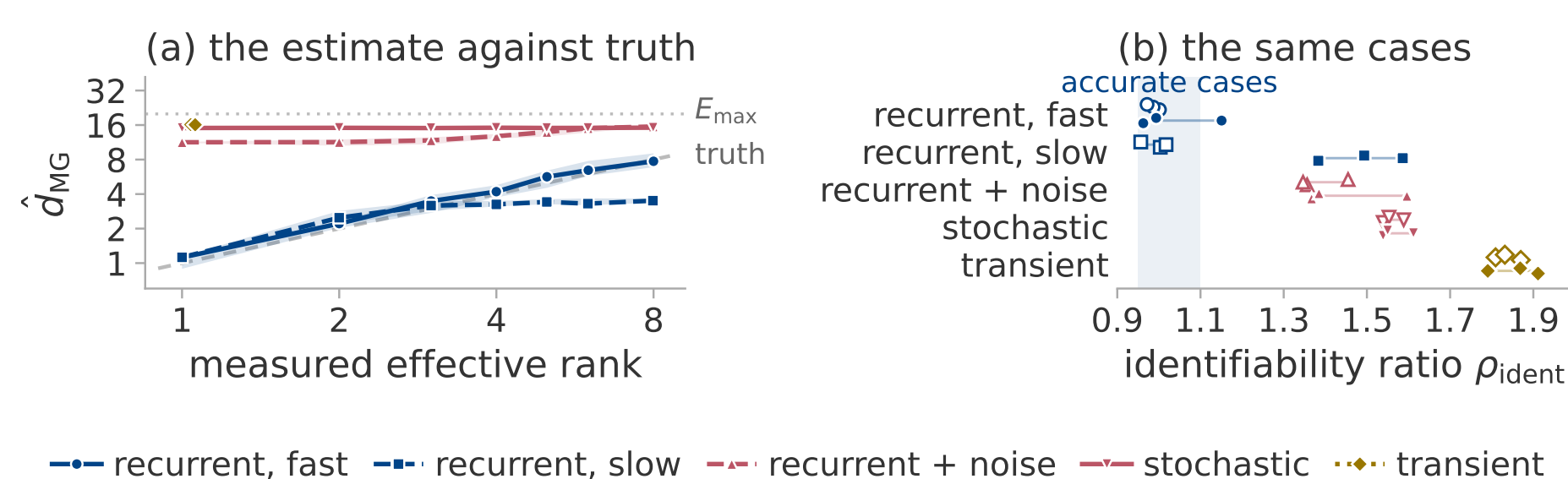
$$\hat{d}_{\text{MG}}^{-1} = \frac{1}{N(m-1)} \sum_{i=1}^N \sum_{j=1}^{m-1} \log \frac{r_j^{(i)}}{r_j^{(i)}}.$$

Pool the nearest-neighbour likelihood over N delay vectors, each with m neighbours, before inversion. $r_j^{(i)}$ is the j th neighbour distance; the **Theiler exclusion** requires time separation greater than W_T .

A dimensional interpretation needs a valid delay embedding, one regime per window, enough returns after exclusion, and a stable scaling range.

03 Controlled recovery has a ceiling

A linear head on a frozen image network is forced by r independent quasiperiodic phases. This supplies known active dimension. Settings are frozen before testing held-out ranks and seeds.



Left: recovery by regime; markers are medians over seeds and observers, bands are interquartile ranges. Right: stability under doubled embedding, at $r = 2$ (open) and $r = 6$ (filled).

- Recurrent:** about one-component error up to roughly eight components at the main configuration.
- Stochastic:** noise shifts the value and makes it depend on embedding settings.
- Transient:** temporal exclusion can produce a large spurious value.

Only one learned-system benchmark passes every prescribed check. All validation systems are externally forced.

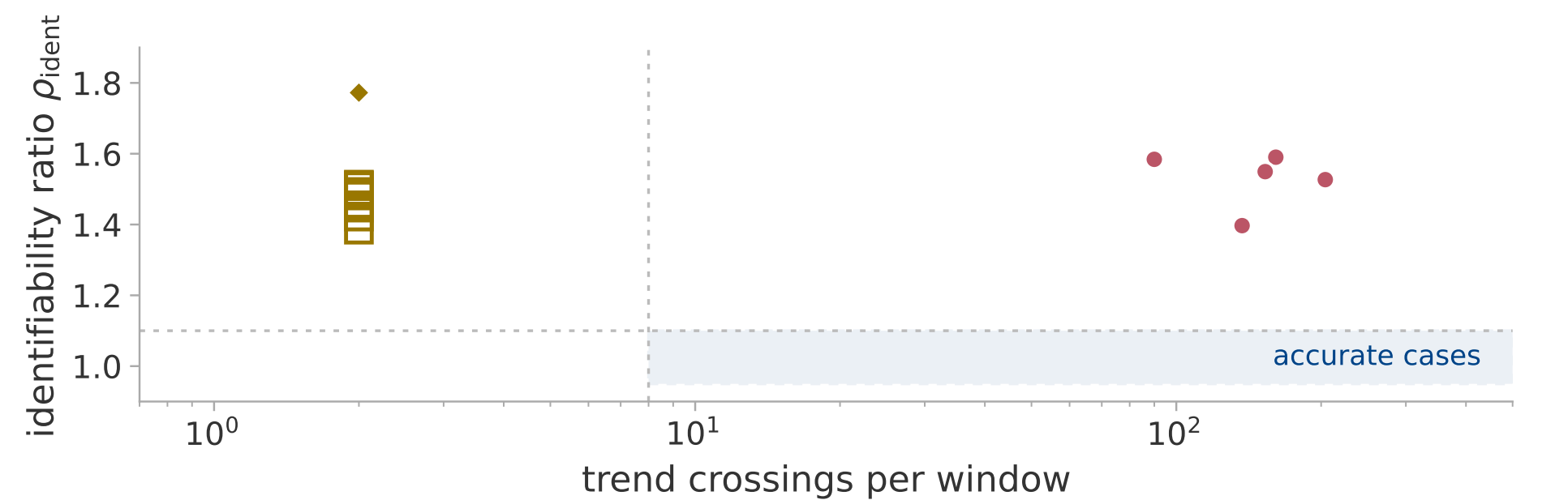
06 Scope of the evidence

- Recovery depends on the lag, observer and window. The tested resolution ceiling is low.
- The grokking comparison has four regularised runs against two without weight decay. Regularisation and outcome are confounded.

Interpret the geometry only after checking the regime. For these grokking runs, the evidence supports a reversible redistribution of trajectory variance.

04 Grokking logs fail the dimension check

Grokking is delayed generalisation after memorisation. We examine a mini-batch transformer and an unregularised full-batch perceptron.



Each point is a log. The blue band records accurate controlled cases; it is not a validated decision boundary.

The **identifiability ratio**

$$\rho_{\text{ident}} = \frac{\hat{d}_{\text{MG}}(2E_{\text{max}})}{\hat{d}_{\text{MG}}(E_{\text{max}})}$$

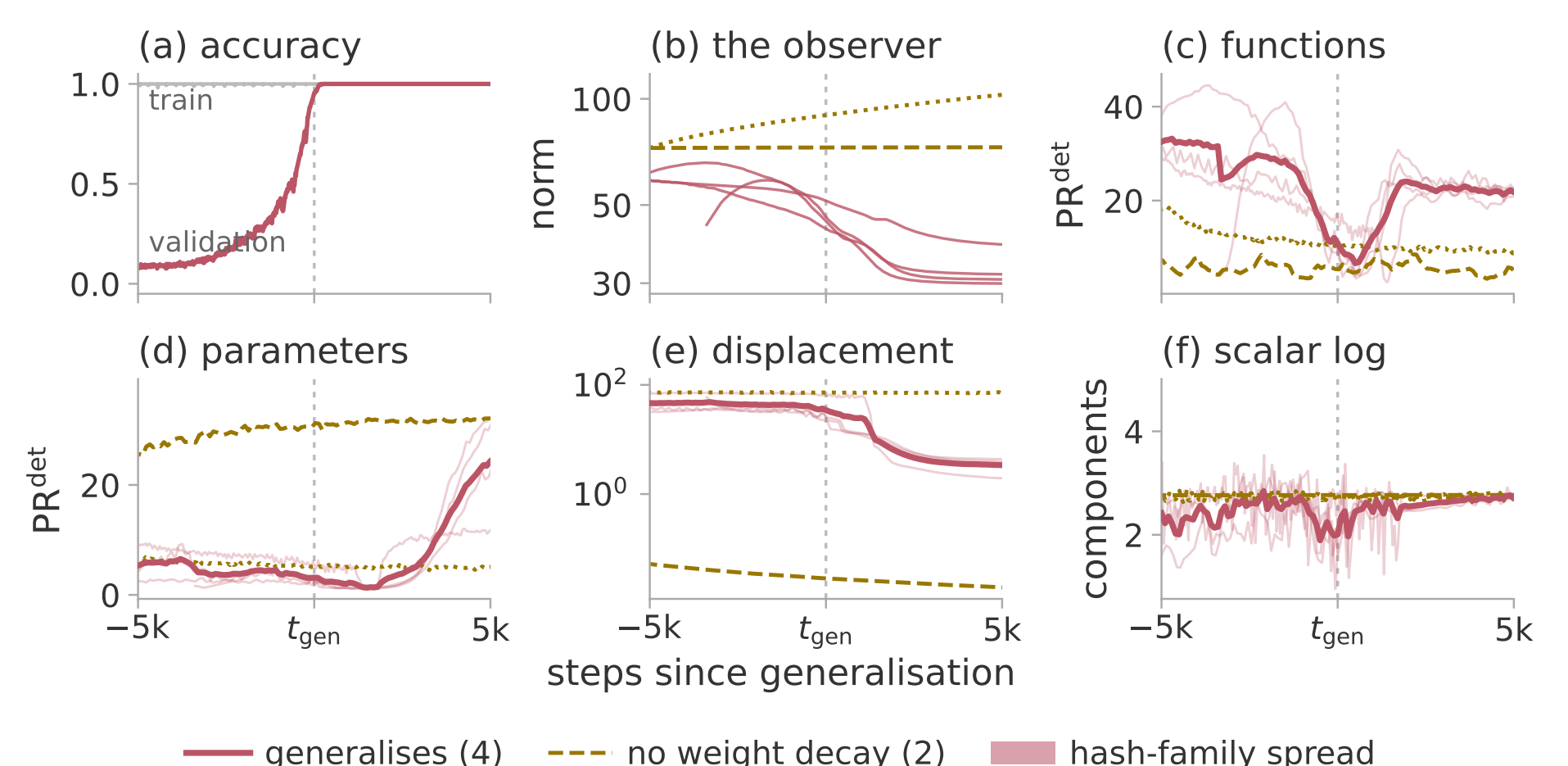
tests sensitivity to embedding dimension. Trend crossings measure non-monotonicity of the log.

All evaluated grokking logs lie outside the reference region. Their scalar levels are not interpreted as active dimension. These diagnostics test neither recurrence itself nor the scaling range.

05 The trajectory rank falls and recovers

We record parameter and probe-logit trajectories through 1024-dimensional CountSketch projections. In each window, detrended covariance eigenvalues λ_i give

$$\text{PR}^{\text{det}} = \frac{(\sum_i \lambda_i)^2}{\sum_i \lambda_i^2}.$$



Four regularised runs (rose): three modular-addition seeds and one S_5 task. Two runs without weight decay (gold) are aligned to paired transition times. Thick rose curves average four runs; shading is hash-family spread, not uncertainty across runs.

- Covariance effective rank falls near generalisation and later re-expands in parameter and function space.
- A window matched to the transition gives a correlated drop in the parameter-norm log estimate.

Effective rank can fall while active dimension stays fixed. The scalar drop describes a transition retrospectively; it does not establish early prediction.

- One memorising run falls just as deeply elsewhere. Depth alone does not distinguish generalisation.
- The scalar effect disappears after linear detrending; roughness changes more strongly. The trajectory result does not reproduce in the full-batch setting.

